

Comparative genome analysis with the human genome reveals chicken genes associated with fatness and body weight.

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The selection of meat-type chickens (broilers) for rapid growth has been accompanied by excessive fat deposition. In this study, we analysed 53 candidate genes that are associated with obesity and obesity-related traits in humans, for which we found chicken orthologues by BLAST searches. We have identified single nucleotide polymorphisms (SNPs) with significant differences in allele frequencies between broilers and layers in each of the following six candidate genes: adrenergic, beta-2-, receptor, surface (ADRB2); melanocortin 5 receptor (MC5R); leptin receptor (LEPR), McKusick-Kaufman syndrome (MKKS), milk fat globule-EGF factor 8 protein (MFGE8) and adenylate kinase 1 (AK1). To examine associations with fatness and/or body weight, we used birds of extreme phenotypes in F(2) and backcross populations with varying levels of abdominal fat weight per cent (%AFW) and body weight. We then assessed the level of gene expression by real-time PCR. In two genes, ADRB2 and MFGE8, we found significant association with %AFW. The ADRB2 gene was found to have a significantly higher expression in the liver of lean chickens compared with those of the fat individuals. We believe that this approach can be applied for the identification of other quantitative genes. © 2011 The Authors, Animal Genetics © 2011 Stichting International Foundation for Animal Genetics.